

Lab 06: OpenMP Parallel Programming

Exercise 01 Report: OpenMP Barrier Directive

1. Exercise Goal

Demonstrate thread synchronization using `#pragma omp barrier`.

2. Source Code

```
#include <omp.h>
#include <stdio.h>

int main() {
    #pragma omp parallel
    {
        int th = omp_get_thread_num();
        printf("thread # %d\n", th);

        #pragma omp barrier

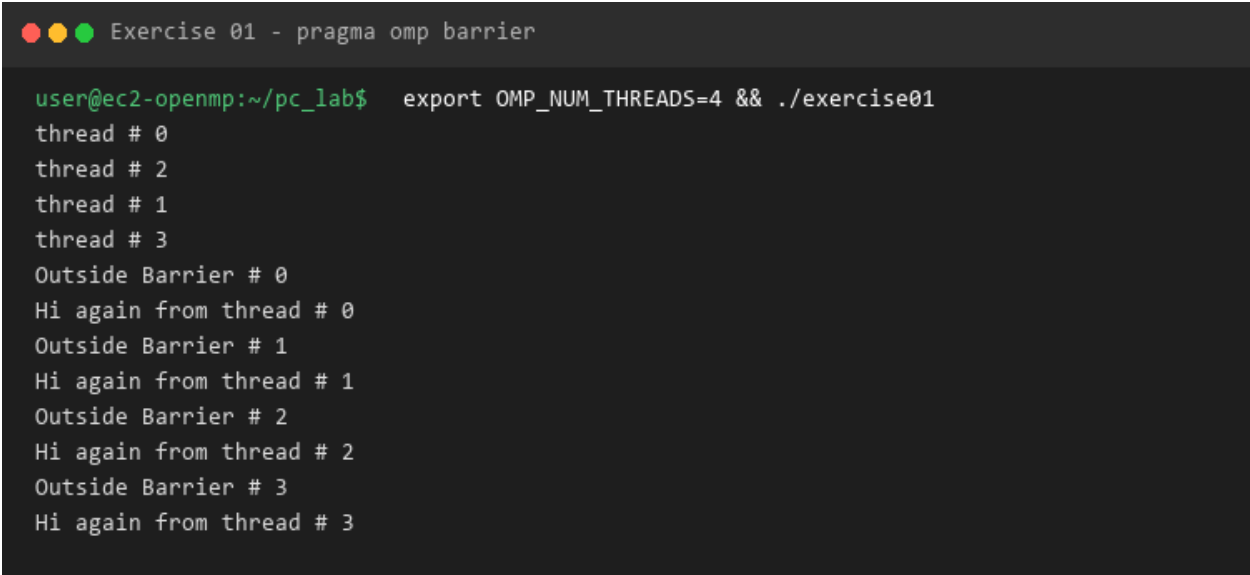
        printf("Outside Barrier # %d\n", omp_get_thread_num());
        printf("Hi again from thread # %d\n", omp_get_thread_num());
    }
    return 0;
}
```

3. Explanation & Analysis

The `#pragma omp barrier` directive acts as a synchronization point for all threads in the parallel team. When execution reaches the barrier, every thread waits until all threads in the team have arrived at the barrier line.

Once the last thread reaches `#pragma omp barrier`, all threads are released at the same time. This guarantees that all pre-barrier thread print statements finish before any post-barrier statement begins.

5. Execution Output Screenshot



```
Exercise 01 - pragma omp barrier

user@ec2-openmp:~/pc_lab$ export OMP_NUM_THREADS=4 && ./exercise01
thread # 0
thread # 2
thread # 1
thread # 3
Outside Barrier # 0
Hi again from thread # 0
Outside Barrier # 1
Hi again from thread # 1
Outside Barrier # 2
Hi again from thread # 2
Outside Barrier # 3
Hi again from thread # 3
```